Md. Zahidul Islam

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Research Interests

Power Electronics · Power Quality Improvement & Harmonic Mitigation · Power Converters · Power System · Smart Grids & Microgrids · Renewable Energy Integration · Machine Learning for Power Systems · Control Systems

Research Experience

Delta/Fork Transformer Based 30-Pulse Star Rectifier for Input Current Harmonic Reduction
Undergraduate Researcher (Thesis)

- Designed and validated a **delta/fork transformer-based 30-pulse star rectifier** in MATLAB/Simulink
- Proposed rectifier comprised two 15-pulse parallelly connected uncontrolled star rectifiers connected to the load through an inter-phase transformer (IPT)
- Achieved **THD <5%** under varying load conditions, complying with **IEEE-519 standards**, demonstrated superior harmonic suppression compared to conventional 6-/12-/18-pulse topologies
- investigated performance in power quality and efficiency for high current and low voltage industrial applications; results published in IEEE conference proceedings

A Soft-Charged Single-Switch Quadratic Boost Converter with Resonant Voltage Lift for High Step-Up Photovoltaic Applications
Ongoing Research

- Designed a soft-charged, single-switch quadratic boost converter with a resonant voltage lift for PV front-ends. It steps 24 V up to 450 V (a gain of about 19) at 200 W and 50 kHz, uses 11 components, has no transformer or coupled inductor, and draws continuous input current
- Derived closed-form expressions for the voltage gain, device voltage stresses, resonant charging current and the zero-current-switching limit. These give a design rule for the resonant inductor ($L_r \leq 1.85\ \mu$H, with zero-current switching held for $D \geq 0.544$)
- Modeled the converter in MATLAB/Simulink and developed a switching-level loss model covering switching, conduction, diode, magnetic and capacitor losses. It predicts 88.9 % efficiency at rated load

Publications

[C1] M. Z. Islam, M. A. Malek, M. A. G. Khan, and M. R. Tahmid, "Delta/Fork Transformer Based 30-Pulse Star Rectifier for Input Current Harmonic Reduction," *4th Int. Conf. on Robotics, Electrical and Signal Processing Techniques (ICREST)*, Dhaka, Bangladesh, 2025, pp. 398–402. | Show Paper

[C2] M. Z. Islam, "A Soft-Charged Single-Switch Quadratic Boost Converter with Resonant Voltage Lift for High Step-Up Photovoltaic Applications," *7th Int. Conf. on Electrical Engineering and Information & Communication Technology (ICEEICT)*, Dhaka, Bangladesh, 2027. *(under review)*

Education

B.Sc. Electrical & Electronic Engineering (EEE) — February 2019– May 2024
Rajshahi University of Engineering & Technology (RUET) — *Bangladesh*

- **CGPA: 3.19/4.00**; final-semester GPA **3.59/4.00** | Certificate
- Core courseworks: Electrical Circuit I & II, Electronics I, II & III, Digital Electronics, Power Electronics, Electrical Machine I & II, Power System I & II, Power Plant Engineering & Economy, Photovoltaic System, Control Systems, Digital Signal Processing, Microprocessors, Interfacing & System Design, Embedded System Design

Professional Experience

Research Consultant — April 2025–Present
Center for Environmental & Geographic Information Services (CEGIS) — *Dhaka, Bangladesh*

- Leading GIS-based electrical modeling of **33/11kV substations** across four zones (Sylhet, Mymensingh, Cumilla, Chattogram) for Bangladesh Power Development Board's project, *GIS-Based Distribution Network System Implementation & 25-Year Masterplan*
- Managed data procurement and validation process for 183 substations across those 4 zones, paid visits to those substations, coordinated with BPDB engineers and field teams to ensure timely delivery of substation data and field reports.
- Inspected and modeled **HV/LV distribution systems, transformers, switchgear, protection & metering systems** for those substations.
- Conducted **load flow studies, loss calculations, fault analysis, and relay coordination** in **ETAP**
- Built utility network basemaps in **ArcMap/ArcGIS Pro**; integrating LT pole and consumer survey data for geospatial analysis
- Prepared technical proposals, monthly reports, and loss-reduction reports with structured findings and identified energy-saving opportunities through loss-reduction analysis – recommended cost-effective solutions

Standardized Test

GRE: 301 (Verbal Reasoning: 145 | Quantitative Reasoning: 156 | Analytical Writing: 3.5)
IELTS: Overall Band Score: 7.0 (Listening: 7.5 | Reading: 7.5 | Writing: 6.5 | Speaking: 6.5)

Technical Skills

Programming	C, C++, MATLAB, Python (learning)
Electrical Simulation	MATLAB/Simulink (Converter modeling), ETAP (Load flow, Relay coordination, Short-circuit analysis) Multisim, PSpice, Proteus, Sprint Layout, TinkerCAD, AutoCAD
Embedded System	Arduino UNO R3, ATmega328, Logo!Soft Comfort, Omron PLC
GIS	ArcMap, ArcGIS Pro (utility network analysis, geospatial data processing)

Selected Academic Projects and Sessional

- **Modeling of DC-DC Converter and H-Bridge Inverter** — Designed a DC to DC converter using PI Controller and H-Bridge Inverter | Report
- **Bidirectional Power Flow between Grid and ESS** — Implemented a Buck-Boost converter for bidirectional power flow between a source and Li-battery | Report
- **Digital Noise Level Detector** — Arduino-based real-time noise monitoring with threshold alerts and custom PCB design | Report
- **Transistor Tester** — Designed and fabricated a standalone parameter tester on a custom PCB board | Report
- **MATLAB GUI App for Cardiac State Detection** — Signal processing app for automated ECG-based cardiac classification | Report
- **AC Motor Drive via DC Source** — Half H-bridge inverter using microcontroller; hardware validated | Report
- **PLC-Based DC Motor Speed Control** — Closed-loop speed regulation via temperature sensor feedback on Omron PLC

Awards & Honors

- **1st Runner-up** — IEEE Student Branch Website Contest, IEEE ISLC 2022 | See more
- **Best Volunteer Award** — IEEE RUET Student Branch (2022–2023) | Certificate
- **National Board Scholarship** — HSC (5.00/5.00, 2018) and SSC (5.00/5.00, 2016)

Certifications & Affiliations

- **IEEE Student Member** #98199339 — Institute of Electrical and Electronics Engineers
- **Hands-on Python: Introduction to Machine Learning** — Workshop by IEEE RUET Student Branch | Certificate

Leadership & Professional Service

Chairperson, IEEE RUET Signal Processing Society (SPS) SB Chapter 2023–2024
IEEE RUET Student Branch
- Led national events, webinars, and workshops on graduate school preparation and signal processing advancements

Chairperson, IEEE RUET Industry Applications Society (IAS) SB Chapter 2022–2023
IEEE RUET Student Branch
- Organized a two-day power systems workshop; arranged a visit to Katakhali 50 MW Peaking Power Plant

Webmaster 2021–2022
IEEE RUET Student Branch
- Monitored and updated the IEEE RUET SB Website, enhancing user experience and accessibility

Joint Secretary 2022–2023
Notre Dame College Association of RUET (NDCAR)

Event Roles — Organizer: RUET EEE Day 2023; Technical Team: ISICT 2022; Logistics: ICECTE 2022; Volunteer: ETECHNOVATION 2K22

Community Engagement

Co-founder 2023–2024
SOPAN (Humanitarian Club), RUET *Bangladesh*
- Led charitable initiatives totaling **≈0.67 million BDT** in community support for underprivileged populations

References

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